Fault-Detection Analytics Pack

Companion figures and verified numbers for the dissertation writer

Regenerated-data era — 2026-11-12

`data/classdetectfault_sim.csv  md5 cf24c7fbe993...  5,365 rows`
`data/detectfault_sim.csv  md5 807df9dd62a4...  9,503 rows`
`branch exp  build 2026-11-12_20-21-33`

Every figure and number in this pack is generated by `src/analytics.py` from re-runnable reports; the traceability appendix maps each one to its source.

1 Dataset and provenance

The pack is built on the Python re-simulation of the corrected 161 kV / 120 km line ($Z_1 = 0.080 + j0.390$, $Z_0 = 0.228 + j1.352$ Ω/km, $C_1 = 12.74$ nF/km). The classification dataset holds 5,365 rows (3,000 fault rows in 375 eight-row location sweeps + 2,365 no-fault rows); the detection dataset holds 9,503 rows and is 31.6% fault rows (45.8% in the superseded original). The scenario-level split assigns 134 of 671 scenarios (1,069 rows) to the shared test partition of all eight models.

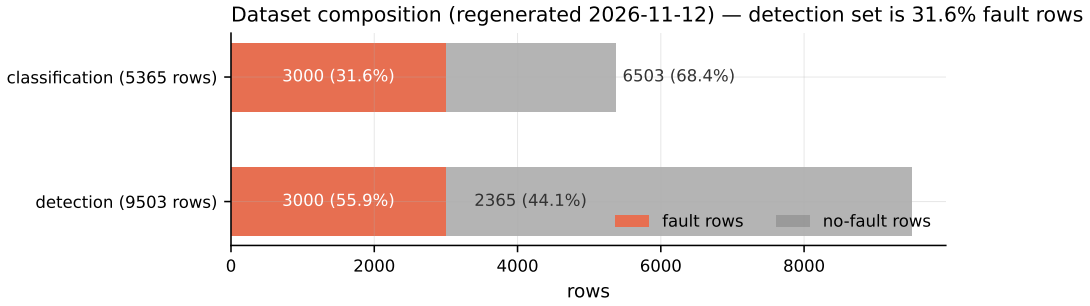


Figure 1: Dataset composition. Where it goes: Section 4.2 (dataset presentation). Source: data files; counts asserted at build time.

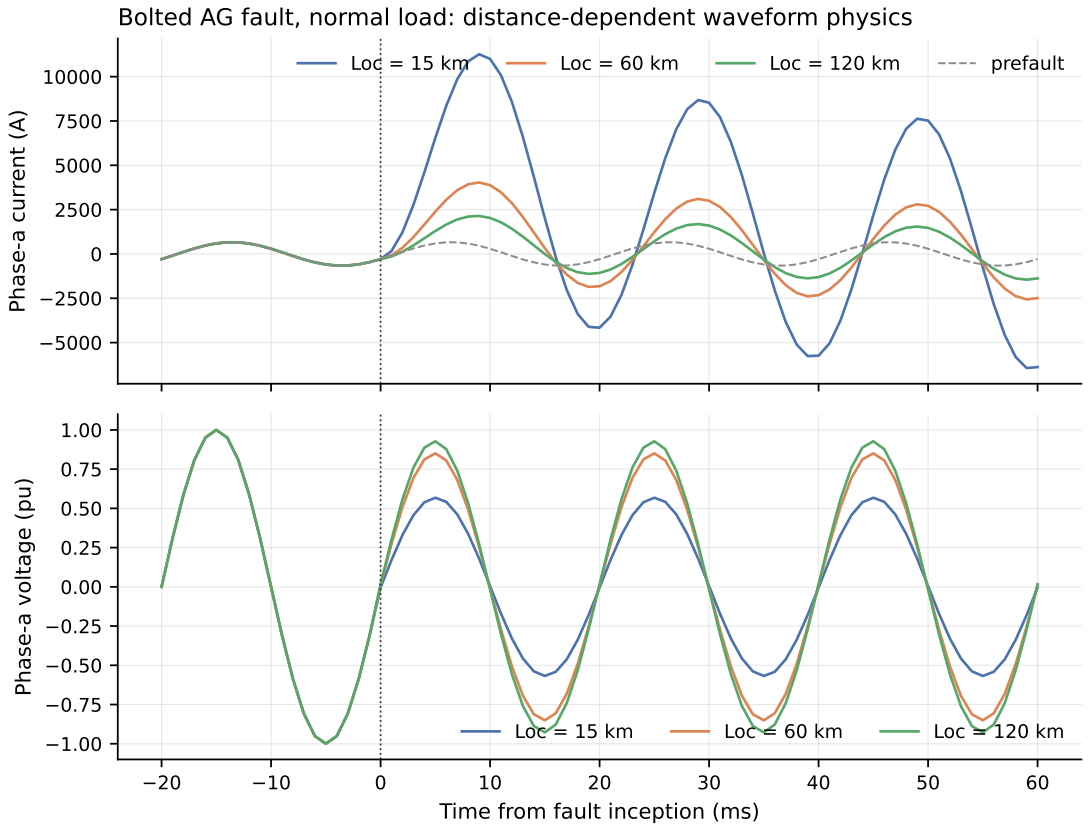


Figure 2: Distance-dependent waveform physics, now present in the regenerated data (the original data was flat in location). Fault current decays 4.76 to 0.89 kA and the faulted-phase voltage sag relaxes 0.568 to 0.928 pu across 15 to 120 km. Where it goes: Section 3.5.5 (dataset generation) and the R10 waveform discussion. Source: src/simulate.py, bolted AG, normal load.

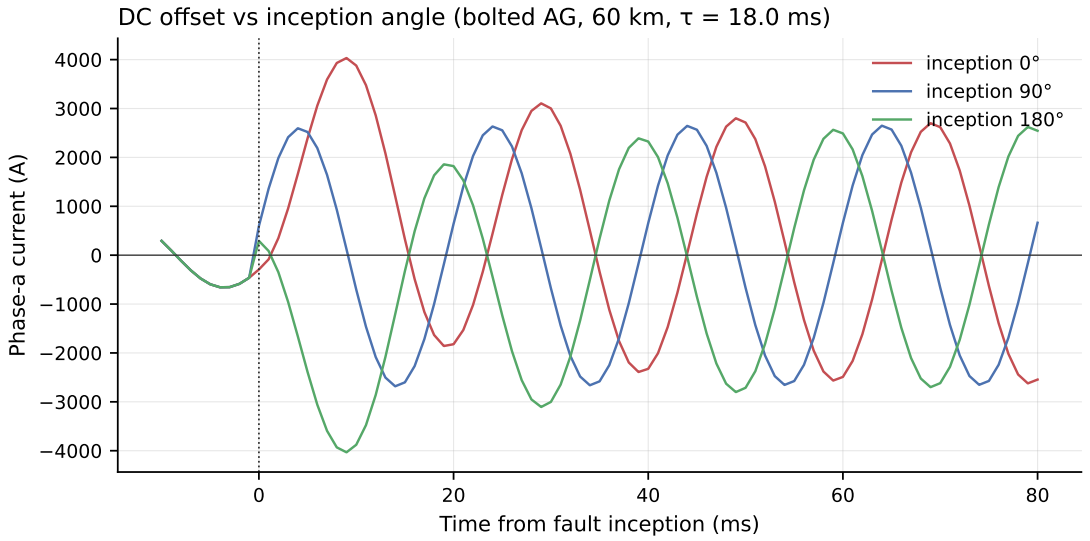


Figure 3: DC offset vs inception angle: the offset peaks at the voltage-zero crossings and vanishes near the voltage peak, decaying with the positive-sequence loop time constant. Where it goes: Section 3.5.5 and R10. Source: src/simulate.py.

2 Conventional relay baseline

The relay is a numerical distance relay deciding on six loop impedances (three residual-compensated ground loops $|V_p|/|I_p + k_0\,3I_0|$, $k_0 = 0.8140 + j0.0405$, and three phase-phase loops) evaluated as steady-state phasor quantities. The Zone 1 reach is $0.80 \times |Z_1| \times 120$ km $= 38.22\ \Omega$. Its profile on the regenerated data: true-negative rate 100%, trip rate 49.2% of fault rows, overall accuracy 0.7158 (all rows) / 0.7568 (common test rows) — secure but incomplete.

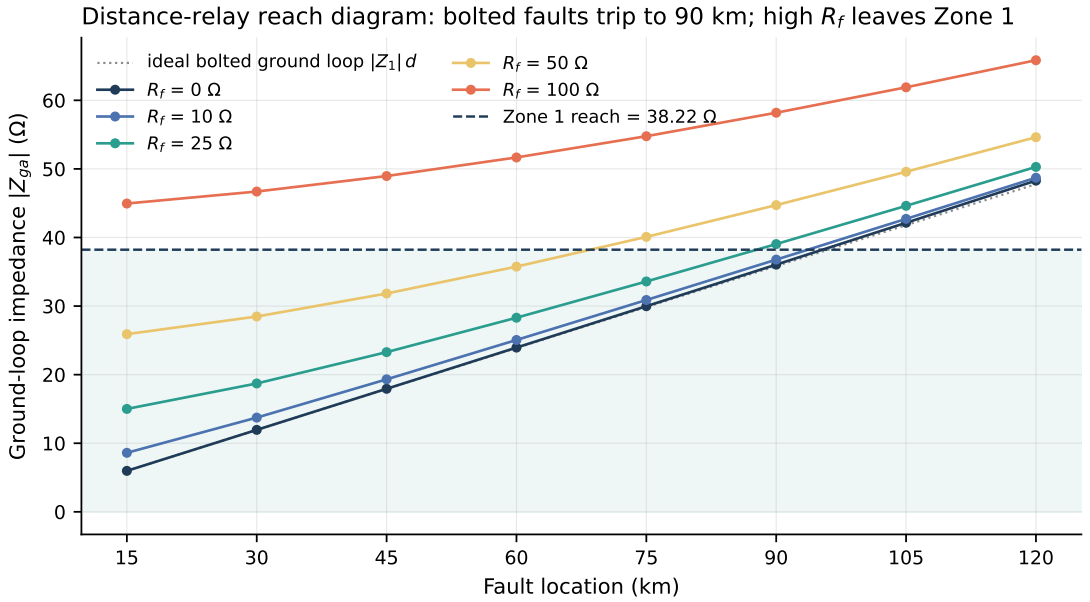


Figure 4: Reach diagram. Bolted faults trip to 90 km ($|Z_1|d = 36.0\ \Omega$ ¡ $38.22\ \Omega$); 105 and 120 km fall beyond Zone 1, and high-resistance faults leave Zone 1 at every location. Where it goes: Section 3.9.4 (relay detection logic). Source: src/simulate.py loop impedances + src/relay.py reach.

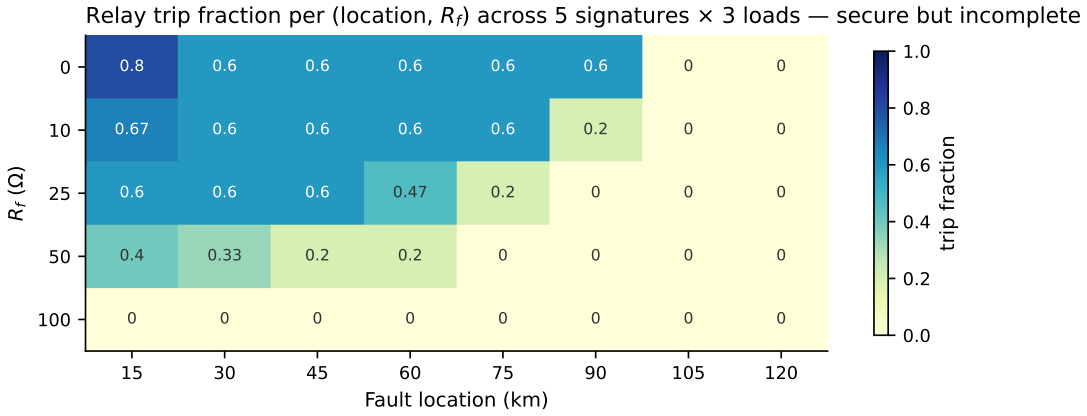


Figure 5: Trip fraction across the full factorial (5 signatures x 3 load levels; the inception angle does not enter the steady-state decision). The white corners — high R_f, far faults — are exactly the regime the learned models cover. Where it goes: Section 4.x, H1 discussion. Source: src/simulate.py.

3 Classification performance (Table 4.3, H1, H2)

All values carry bootstrap SDs (2,000 resamples, seed 42) on the 1,069 common test rows. The weakest cell in the table is the RNN on phase A (accuracy 0.9280); the LSTM attains 1.0 on every target.

model	n	accuracy	recall	F1	robustness	latency (ms)
ann	1069	0.9944 ± 0.0024	0.9900 ± 0.0041	0.9950 ± 0.0020	0.9836 ± 0.0009	0.0011 ± 0.0002
cnn	1069	0.9981 ± 0.0014	0.9967 ± 0.0023	0.9983 ± 0.0012	0.9853 ± 0.0006	0.0557 ± 0.0070
dtr	1069	0.9972 ± 0.0016	0.9950 ± 0.0029	0.9975 ± 0.0015	0.9383 ± 0.0046	0.0007 ± 0.0001
knn	1069	0.9963 ± 0.0019	0.9933 ± 0.0034	0.9967 ± 0.0017	0.9819 ± 0.0009	0.0148 ± 0.0027
lstm	1069	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000	204.91 ± 103.85
pinn	1069	0.9972 ± 0.0016	0.9950 ± 0.0028	0.9975 ± 0.0014	0.9840 ± 0.0011	0.0038 ± 0.0005
rnn	1069	0.9935 ± 0.0024	0.9883 ± 0.0044	0.9941 ± 0.0023	0.9842 ± 0.0011	0.0008 ± 0.0000
svm	1069	0.9991 ± 0.0009	0.9983 ± 0.0016	0.9992 ± 0.0009	0.9804 ± 0.0010	0.0011 ± 0.0002
relay	1069	0.7568 ± 0.0130	0.5667 ± 0.0201	0.7234 ± 0.0165	n/a	10.00 ± 0.00
relay (all rows)	5365	0.7158 ± 0.0063	0.4917 ± 0.0091	0.6592 ± 0.0082	n/a	10.00 ± 0.00

Table 1: Table 4.3 (regenerated), overall rows. Relay "all rows" covers the full 5,365-row dataset. Source: outputs/table_4_3 (report 2026-11-12_19-43-50).

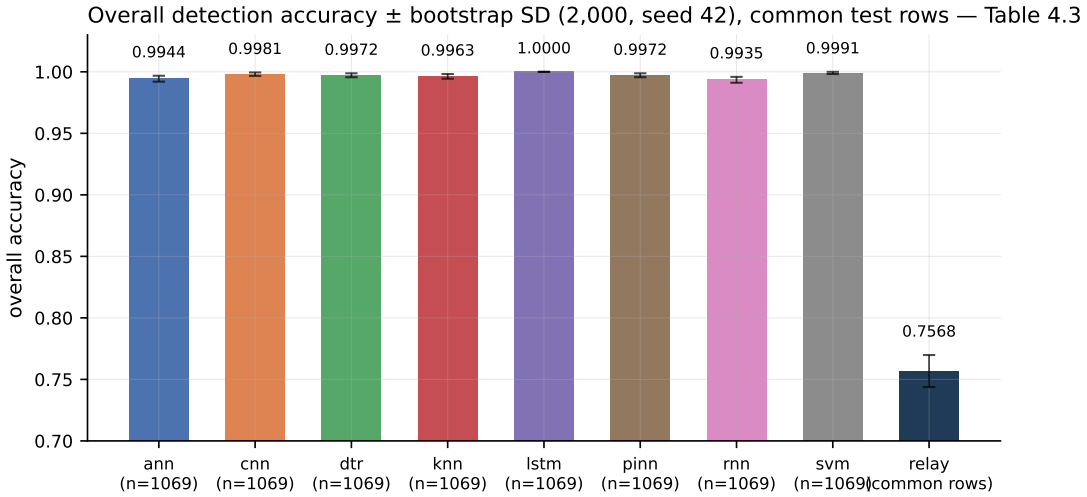


Figure 6: Overall detection accuracy with bootstrap SDs. Where it goes: Section 4.x, Table 4.3 discussion. Source: table_4_3.csv.

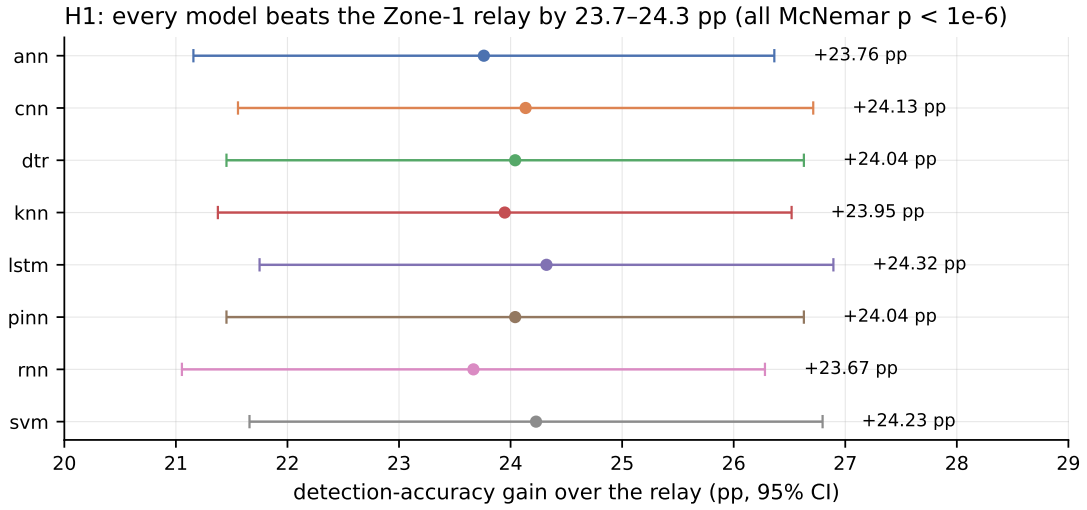


Figure 7: H1 forest plot: accuracy gain over the relay with 95% CIs; Cochran's Q = 1906.85 (df = 8, $p < 10^{-6}$). Where it goes: Section 4.x, H1. Source: hypothesis results CSV.

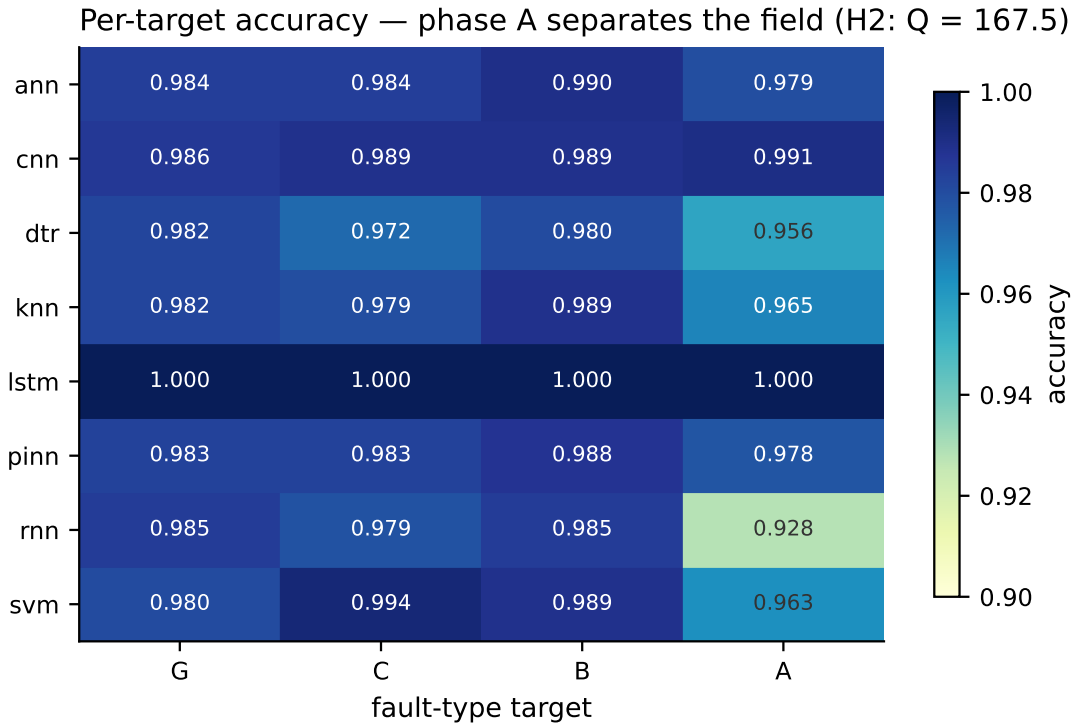


Figure 8: Per-target accuracy. Per-target Cochran's Q: G: Q = 40.04, C: Q = 60.13, B: Q = 37.68, A: Q = 167.49; under Holm correction the LSTM is significantly best on G/C/B and, with CNN, on phase A. Where it goes: Section 4.x, H2. Source: hypothesis results CSV.

4 Latency and noise robustness (H3, H4)

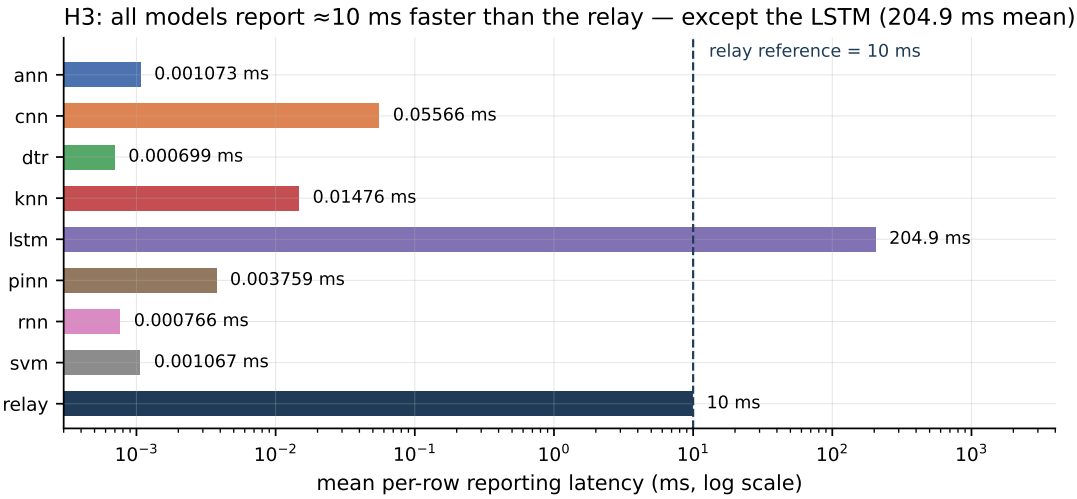


Figure 9: Reporting latency (log scale). Every model reports faster than the relay's 10 ms reference by essentially the full margin (Hodges–Lehmann ≈ −10 ms); the LSTM reports 188.75 ms slower (95% CI [178.5, 197.0], mean 204.9 ms). Where it goes: Section 4.x, H3 — replaces the stale "270 ms" sentence. Source: table_4_3.csv + hypothesis results CSV.

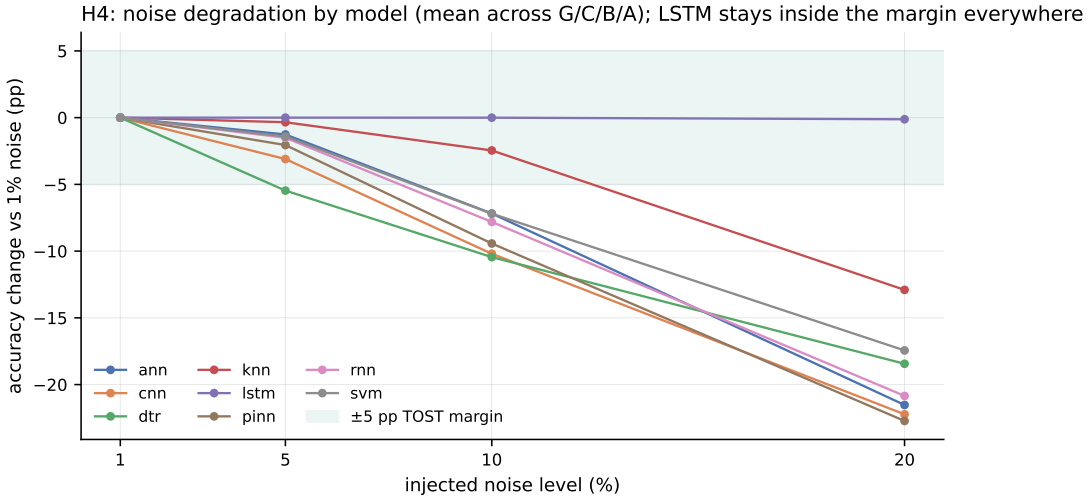


Figure 10: Accuracy change vs the 1% noise baseline, mean across targets. The shaded band is the pre-declared 5 pp TOST equivalence margin: the LSTM is equivalent at every level, KNN up to 10% on all targets, the others break at 20%; CNN and PINN are not equivalent on phase A even at 5%. Where it goes: Section 4.x, H4. Source: robustness_*.csv (2026-11-12 rows).

5 Fault location and scenario structure (H5, R9)

Location accuracy is measured on the 600 fault scenarios of the common test set (MAE as % of the 120 km line).

model	locations	MAE (km)	MAE (% of 120 km)
lstm	8	0.46	0.380
dtr	8	2.23	1.858
knn	8	5.18	4.319
cnn	8	9.68	8.066
ann	8	13.54	11.282
svm	8	13.61	11.345
rnn	8	15.23	12.692
pinn	8	15.88	13.237

Table 2: H5 fault-location MAE, recomputed from the Loc columns of the prediction files and asserted against the hypothesis report (within 0.05 pp). Source: predictions_*.csv.

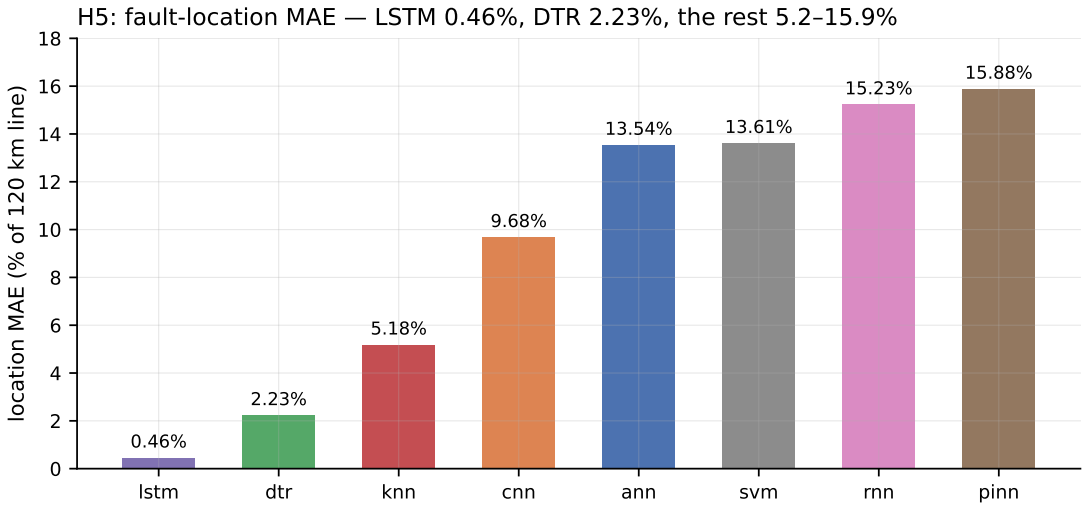


Figure 11: Location MAE ranking. Where it goes: Section 4.x, H5. Source: prediction files (asserted).

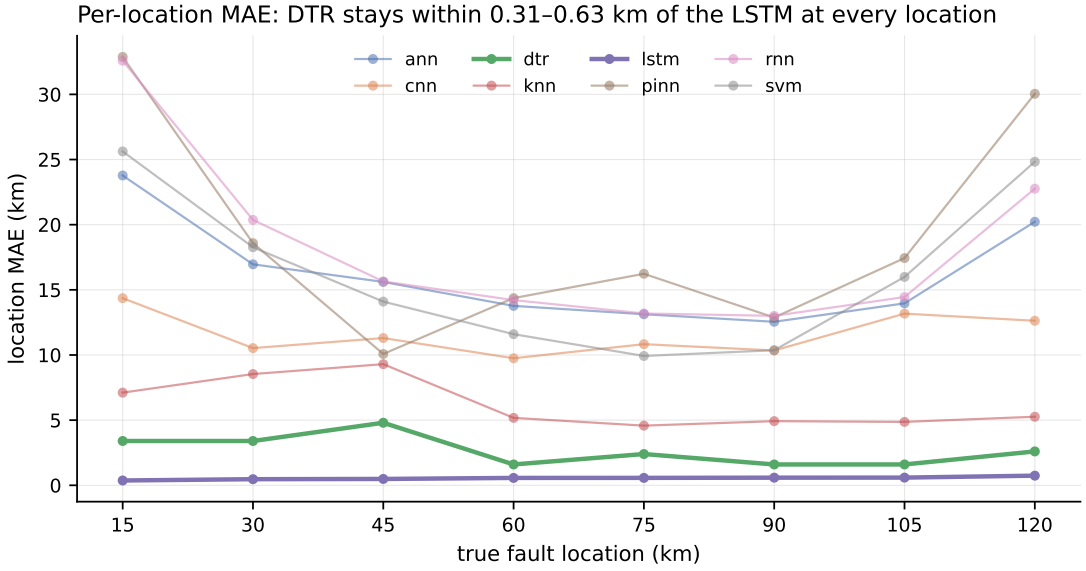


Figure 12: Per-location MAE. The DTR stays within 0.31–0.63 km of the LSTM at every location (all Holm-adjusted $p < 10^{-6}$); every other model trails by 3.3–30.9 km. Friedman all-locations $\chi^2 = 2338.9$, Kendall's W = 0.557. Where it goes: Section 4.x, H5. Source: prediction files.

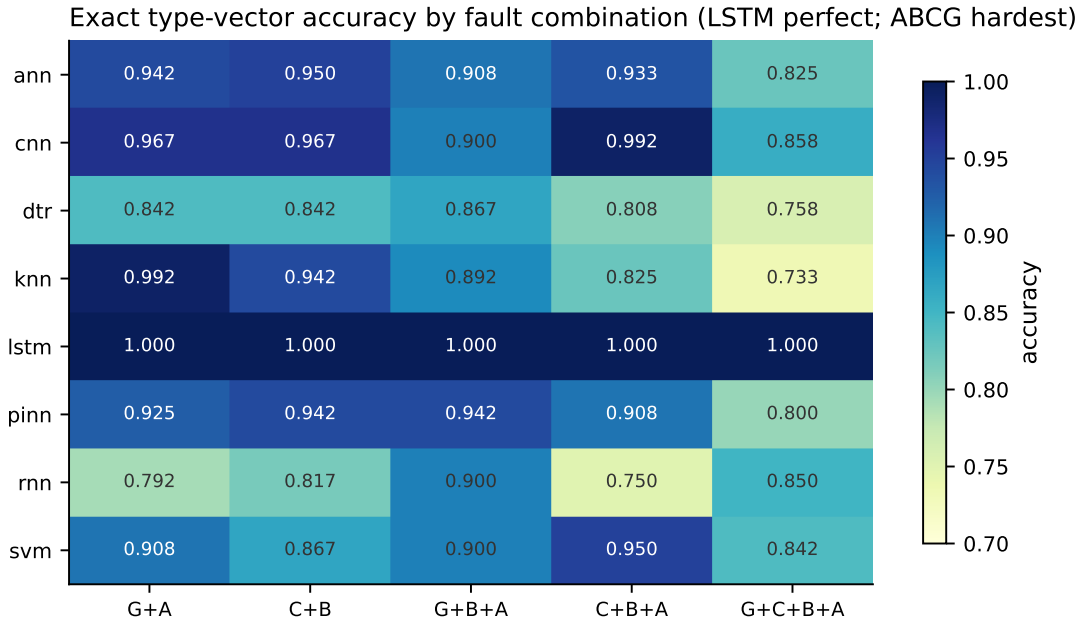


Figure 13: Exact type-vector accuracy by fault combination; the relay row is structurally inapplicable (binary trip output) and is omitted. Where it goes: Section 4.x, combination-accuracy discussion. Source: combination results CSV.

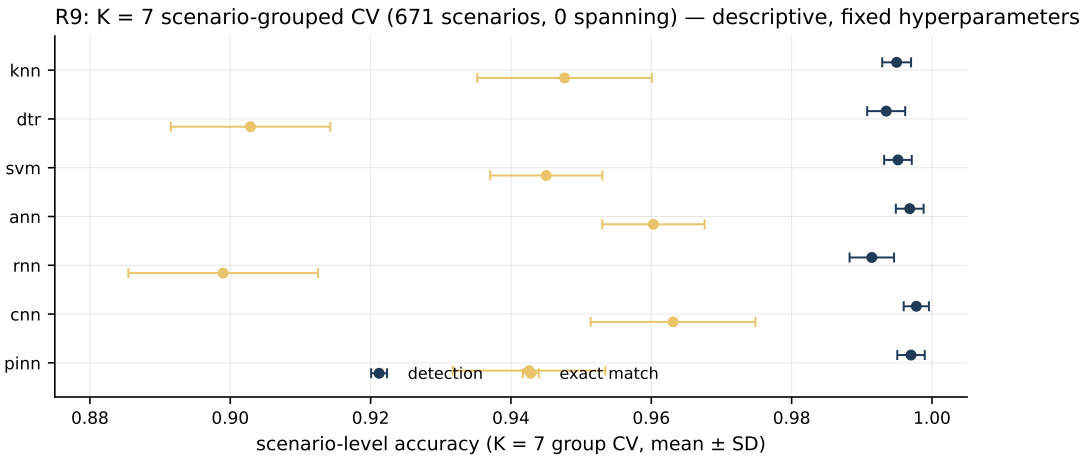


Figure 14: Scenario-grouped K = 7 CV (descriptive, fixed hyperparameters): detection 0.9914–0.9978, exact match 0.8990–0.9631. Where it goes: Section 3.12.1 / 4.x (R9). Source: cv_results CSV.

model	detection acc.	exact-match acc.
knn	0.9950 ± 0.0021	0.9476 ± 0.0124
dtr	0.9935 ± 0.0027	0.9029 ± 0.0114
svm	0.9952 ± 0.0020	0.9450 ± 0.0080
ann	0.9968 ± 0.0020	0.9603 ± 0.0073
rnn	0.9914 ± 0.0032	0.8990 ± 0.0135
cnn	0.9978 ± 0.0018	0.9631 ± 0.0117
pinn	0.9970 ± 0.0020	0.9426 ± 0.0109

Table 3: CV means ± SD across the 7 scenario-grouped folds. Source: outputs/scenario_cv (report 2026-11-12_17-25-03).

A scenario-level leakage audit found zero spanning scenarios for all nine methods (models touch 134 of 671 test scenarios; the relay, which trains on nothing, covers all 671).

6 Writer's companion

6.1 Key takeaways, keyed to the chapter

- **§3.3.1** — correct the line-charging figure to 4.15 MVAr (120 km section) / 10.28 MVAr (297.4 km route); keep the 10.8 kA bolted three-phase cross-check.
- **§3.3** — insert the declared auxiliary assumptions (X/R = 10; 0.95 pf constant-impedance load at 40/75/100% of 750 A; $C_0 = 0.75\,C_1$; Z_0 source = Z_1 source; bolted ground; $R_f = 0 \equiv 1$ mΩ; EMF calibration) and the load-bus operating points 0.931/0.862/0.815 pu.
- **§3.5.5** — new design space: 375 sweeps x 8 locations = 3,000 fault rows; datasets 5,365 / 9,503 rows; detection set 31.6% fault rows; event-sweep model caveat.
- **§3.8.2/3.12.1** — 671 scenarios, shared scenario-level split (134 test scenarios / 1,069 rows), K = 7 CV is descriptive only.
- **§3.9.4** — relay measurement chain: six loops, complex k_0, Zone 1 = 38.22 Ω, steady-state phasor semantics; secure-but-incomplete profile (TN 100%, TP 49.2%).
- **§4.x** — Table 4.3, H1–H5 numbers from this pack; LSTM latency gap is ≈189 ms, not 270 ms.

6.2 Do NOT transcribe (superseded numbers)

- "12,000 scenarios / 14,400 rows / 2,400 held-out scenarios / 7,861 rows"; "45.8% fault rows"; "5,496 fault rows / 685 sweeps + 4 truncated".
- "24,000 fault rows" (design-space double-count).
- "≈30.9 MVAr line charging" ($\sqrt{3}$ slip).
- "LSTM latency gap ≈ 270 ms".
- Any accuracy, p-value, latency or table cell computed from the original datasets.
- Old relay narrative ("0.9+ detection accuracy").
- Old absolute waveform claims (44 A no-fault currents, 4% of nominal faulted-phase voltage, 0.62 prefault peak).

6.3 Traceability appendix

tag	source
SIM	`src/simulate.py` imported directly; 8-point verification battery all PASS (outputs/simulate/report_2026-11-12_16-09-04.txt); datasets md5-identical across runs
REL	outputs/relay/report_2026-11-12_16-20-30.txt
T43	outputs/table_4_3/report_2026-11-12_19-43-50.txt
HYP	outputs/hypothesis_tests/report_2026-11-12_17-08-01.txt
CV	outputs/scenario_cv/report_2026-11-12_17-25-03.txt
COMB	outputs/combination_accuracy/report_2026-11-12_17-25-01.txt
LEAK	outputs/leakage_audit/report_2026-11-12_17-25-01.txt
ROB	outputs/<model>/robustness_<model>.csv (2026-11-12 rows)